

Deque - Introduction and Applications

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Deque or Double Ended Queue is a generalized version of [Queue data structure](#) that **allows insert and delete at both ends**. Below is an example program of deque in different languages.

- Deque can act as both Stack and Queue
- It is useful in many problems where we need to have a subset of all operations also like insert/remove at front and insert/remove at the end.
- It is typically implemented either using a doubly linked list or circular array.

Implementations of Deque in Different Languages

- [deque in C++](#)
- [Deque in Java](#)
- [deque in Python](#)
- [Deque in JavaScript](#)

Below are example programs in different languages.

C++

Java

Python

C#

```
#include <iostream>
#include <deque>

using namespace std;

int main() {
    deque<int> dq;

    dq.push_back(10);
    dq.push_back(20);
    dq.push_front(30);

    // Print deque elements
    for (int x : dq) cout << x << " ";
    cout << endl;
```

```
// Pop from front and back
dq.pop_front();
dq.pop_back();

// Print deque elements after pop
for (int x : dq) cout << x << " ";

return 0;
}
```

Deque Basics

- [Array Implementation of Deque](#)
- [Linked List Implementation of Deque](#)

Practice Problems Based on Deque

Basic Problems

- [Difference between Queue and Deque](#)
- [Deque Implementation in Python](#)
- [First and Last Elements of Deque in Python](#)
- [Add Element at Front of a Deque](#)
- [Remove an Element from Front of Deque](#)
- [Minimize Maximum Difference Between Adjacent Elements](#)

Easy Problems

- [Rearrange Linked List to Alternate First and Last](#)
- [Substring with Maximum Frequency](#)
- [Prefixes as Suffixes of a String](#)
- [Level order traversal in spiral form](#)
- [String after processing backspace characters](#)
- [Generate a Sequence by inserting positions](#)
- [Lexicographically largest permutation](#)
- [Check if Strings Can Be Made Equal](#)

Medium Problems

- [Stack and Queue Using ArrayDeque in Java](#)
- [Implement Stack and Queue using Deque](#)
- [Generate Bitonic Sequence](#)
- [Rearrange Array Elements](#)
- [Longest Subarray with Absolute Difference \$\leq X\$](#)
- [Reverse a Linked List in groups](#)
- [Max Sum Subsequence with K Distant Elements](#)
- [Nth term of given recurrence relation](#)
- [Max Subarray Length with K Increments](#)
- [Largest String after Deleting K Characters](#)
- [Segregate even and odd nodes in a Linked List](#)
- [Generate Permutation with Unique Adjacent Differences](#)
- [0-1 BFS](#)
- [Min Deques to Sort Array](#)
- [Min Number by Applying + and * Operations](#)

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